Decision Tree

Tennis Playing

- Tennis playing record

	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Tennis Playing

- Today is Sunny, Mild, High and Strong
  - Will he play tennis today?
- How can you predict his tennis playing?
- What is the rule for playing tennis?
  - How to extract patterns from the data?

Basic Idea

- We have a simple data

Day	A	B	C	Play
1	T	T	T	T
2	T	T	F	T
3	F	T	F	T
4	F	F	T	F
5	F	F	F	F

  - Will he play tennis if A=F, B=T, C=T ?
  - Will he play tennis if A=T, B=F, C=F ?

Basic Idea

- A simple idea: See one of inputs and answer based on the majority
  - Will he play tennis if A=F, B=T, C=T ?
  - Will he play tennis if A=T, B=F, C=F ?

Day	A	B	C	Play
1	T			T
2	T			T
3	F			T
4	F			F
5	F			F

If A = T then {T,T}
else {T,F,F}

Day	A	B	C	Play
1		T		T
2		T		T
3		T		T
4		F		F
5		F		F

If B = T then {T,T,T}
else {F,F}

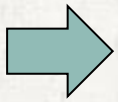
Day	A	B	C	Play
1			T	T
2			F	T
3			F	T
4			T	F
5			F	F

If C = T then {T,F}
else {T,T,F}

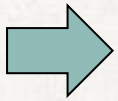
Basic Idea

- Conclusion

Day	A	B	C	Play
1	T	T	T	T
2	T	T	F	T
3	F	T	F	T
4	F	F	T	F
5	F	F	F	F



If B = T then
{T,T,T}
else
{F,F}



If B = T then
Play = T
else
Play = F

Why splitting with *B* is the best?

– *B* splits *Play* into homogeneous subsets

- We can extend this idea

Entropy

- We have 4 sets of 10 data
  - Data Set1: 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
  - Data Set2: 0, 0, 0, 0, 0, 0, 0, 0, 0, 0

  ⇒ Are the sets disordered? (Heterogeneity)

  - Data Set3: 1, 1, 1, 1, 1, 0, 0, 0, 0, 0
  - Data Set4: 1, 1, 1, 0, 0, 0, 0, 0, 0, 0

  ⇒ Which set is more disordered than the other?

Entropy

- A function measuring the degree of disorder (heterogeneity)

Set1: 1, 1, 1, 1, 1, 1, 1, 1
Set2: 1, 1, 1, 1, 1, 1, 1, 0
Set3: 1, 1, 1, 1, 1, 1, 0, 0
Set4: 1, 1, 1, 1, 1, 0, 0, 0
Set5: 1, 1, 1, 1, 0, 0, 0, 0
Set6: 1, 1, 1, 0, 0, 0, 0, 0
Set7: 1, 1, 0, 0, 0, 0, 0, 0
Set8: 1, 0, 0, 0, 0, 0, 0, 0
Set9: 0, 0, 0, 0, 0, 0, 0, 0

Entropy

- A function measuring the degree of disorder (heterogeneity)

$$Entropy(set) = -P_1 \log_2 P_1 - P_0 \log_2 P_0$$

P_1 = the probability that 1 appears in *set*

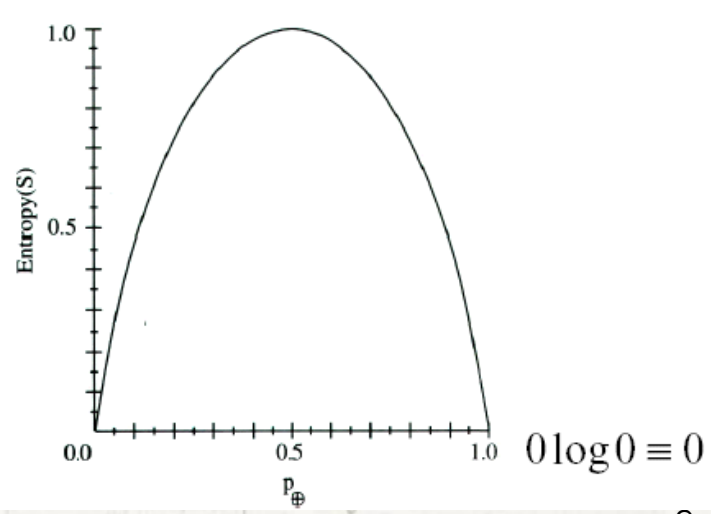
P_0 = the probability that 0 appears in *set*

A={1,1,1,0,0,0,0,0}

$$Entropy(A) = -P_1 \log_2 P_1 - P_0 \log_2 P_1$$
$$= 0.954$$

$P_1 = 3/8$

$P_0 = 5/8$



Entropy

- A function measuring the degree of disorder
  - If *set* is composed of $c_1, c_2, \ldots, c_n$

$$Entropy(set) = -\sum_{i=1}^{n} P_{c_i} \log_2 P_{c_i}$$

$$P_{c_i} = \text{the probability that } c_i \text{ appears in } set$$

Entropy

- Entropy of Partitions
  - Average of entropies of each partition

$$E(S,P)=\sum_{p\in P}\frac{|p|}{|S|}Entropy(p)$$

  - Example: which partitioning is better?

1, 1, 1, 1, 0, 0, 0, 0

$$P_1=\{\{1,1,0,0,0\},\{1,1,0\}\}$$

$$P_2=\{\{1,1,1,1,0\},\{0,0,0\}\}$$

Entropy

- Entropy of Partitions
  - Example:

Partitioning 1

1, 1, 0, 0, 0

$$-P_1\log_2 P_1 - P_0\log_2 P_0 = -2/5\log_2 2/5 - 3/5\log_2 3/5 = 0.97$$

1, 1, 0

$$-P_1\log_2 P_1 - P_0\log_2 P_0 = -2/3\log_2 2/3 - 1/3\log_2 1/3 = 0.92$$

$$\frac{5}{8}\times 0.97 + \frac{3}{8}\times 0.92 = 0.95$$

Partitioning 2

1, 1, 1, 1, 0

$$-P_1\log_2 P_1 - P_0\log_2 P_0 = -4/5\log_2 4/5 - 1/5\log_2 1/5 = 0.72$$

0, 0, 0

$$-P_1\log_2 P_1 - P_0\log_2 P_0 = -3/3\log_2 3/3 - 0/3\log_2 0/3 = 0.0$$

$$\frac{5}{8}\times 0.81 + \frac{3}{8}\times 0 = 0.51$$

Selecting Attribute

- Let's return back to a simple data

Day	A	B	C	Play
1	T	T	T	T
2	T	T	F	T
3	F	T	F	T
4	F	F	T	F
5	F	F	F	F

Play:{3T, 2F}

- Which attribute is good for decision of "Play"
  - B, why?

If A=T {2T,0F}
If A=F {1T,2F}

If B=T {3T,0F}
If B=F {0T,2F}

If C=T {1T,1F}
If C=F {2T,1F}

- Which attribute partitions "Play" with the lowest entropy?

Selecting Attribute

Day	A	B	C	Play
1	T	T	T	T
2	T	T	F	T
3	F	T	F	T
4	F	F	T	F
5	F	F	F	F

- Let's return back to a simple data
  - Before partitioning

Play:{3T, 2F} $-P_T \log_2 P_T - P_F \log_2 P_F = -3/5 \log_2 3/5 - 2/5 \log_2 2/5 =$ 0.97

  - Which attribute partitions "Play" with the lowest entropy?

$\text{Play}_{A=T}$:{2T,0F} $-P_T \log_2 P_T - P_F \log_2 P_F = -1.0 \log_2 1.0 - 0.0 \log_2 0.0 = 0.0$

$\text{Play}_{A=F}$:{1T,2F} $-P_T \log_2 P_T - P_F \log_2 P_F = -1/3 \log_2 1/3 - 2/3 \log_2 2/3 = 0.92$

$$\frac{2}{5} \times 0 + \frac{3}{5} \times 0.92 = 0.55$$

$\text{Play}_{B=T}$:{3T,0F} $-P_T \log_2 P_T - P_F \log_2 P_F = -1.0 \log_2 1.0 - 0.0 \log_2 0.0 = 0.0$

$\text{Play}_{B=F}$:{0T,2F} $-P_T \log_2 P_T - P_F \log_2 P_F = -1.0 \log_2 1.0 - 0.0 \log_2 0.0 = 0.0$

$$\frac{3}{5} \times 0.0 + \frac{2}{5} \times 0.0 = 0.0$$

$\text{Play}_{C=T}$:{1T,1F} $-P_T \log_2 P_T - P_F \log_2 P_F = -1/2 \log_2 1/2 - 1/2 \log_2 1/2 = 1.0$

$\text{Play}_{C=F}$:{2T,1F} $-P_T \log_2 P_T - P_F \log_2 P_F = -2/3 \log_2 2/3 - 1/3 \log_2 1/3 = 0.92$

$$\frac{2}{5} \times 1.0 + \frac{3}{5} \times 0.92 = 0.95$$

Selecting Attribute

Day	A	B	Play
1	T	T	T
2	T	T	T
3	F	T	T
4	F	T	F
5	F	F	F

- Another example
  - Which attribute partitions "Play" with the lowest entropy?

Day	A	B	Play
1	T		T
2	T		T
3	F		T
4	F		F
5	F		F

If A=T: {T,T}

$$-P_T \log_2 P_T - P_F \log_2 P_F = -1.0 \log_2 1.0 - 0.0 \log_2 0.0 = 0.0$$

If A=F: {T, F, F}

$$-P_T \log_2 P_T - P_F \log_2 P_F = -2/3 \log_2 2/3 - 1/3 \log_2 1/3 = 0.92$$

$$\frac{2}{5} \times 0 + \frac{3}{5} \times 0.92 = 0.55$$

Day	A	B	Play
1		T	T
2		T	T
3		T	T
4		T	F
5		F	F

If B=T: {T,T,T,F}

$$-P_T \log_2 P_T - P_F \log_2 P_F = -3/4 \log_2 3/4 - 1/4 \log_2 1/4 = 0.81$$

If B=F: {F}

$$-P_T \log_2 P_T - P_F \log_2 P_F = -0 \log_2 0 - 1 \log_2 1 = 0.0$$

$$\frac{4}{5} \times 0.81 + \frac{1}{5} \times 0 = 0.44$$

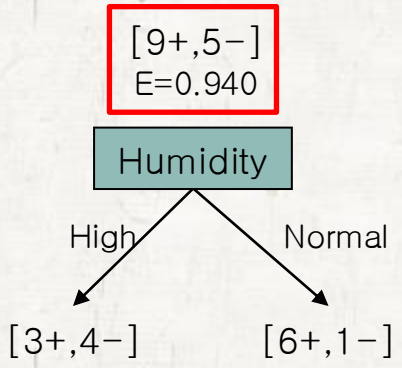
Selecting Attribute

- Let's return to Tennis Data. Which attribute will divided Yes and No well enough?
  - An attribute which the most reduces the degree of disorder
  - How to measure how much entropy is reduced by an attribute?

Example

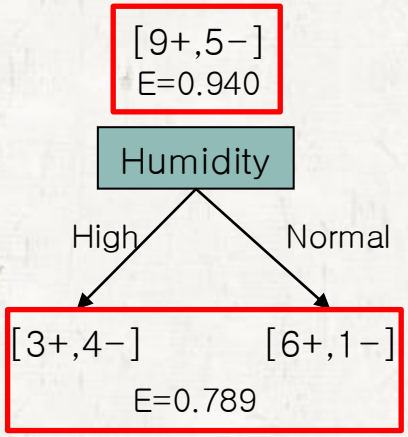
- Tennis playing record again

	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No



Selecting Attribute

- Reduction of disorder after dividing
  - S is divided into two subsets
  - What is the average entropy of the two subsets ?



Entropy of the subset with Humidity = High

$$Entropy(S_{Humidity=High}) = -\frac{3}{7}\log_2\frac{3}{7} - \frac{4}{7}\log_2\frac{4}{7} = 0.985$$

Entropy of the subset with Humidity = Normal

$$Entropy(S_{Humidity=Normal}) = -\frac{6}{7}\log_2\frac{6}{7} - \frac{1}{7}\log_2\frac{1}{7} = 0.592$$

Average entropy of the two subset

$$\frac{7}{14}\times 0.985 + \frac{7}{14}\times 0.592 = 0.789$$

Selecting Attribute

- Reduction of disorder after dividing
  - Entropy before dividing with Humidity : 0.940
  - Entropy after dividing with Humidity : 0.789

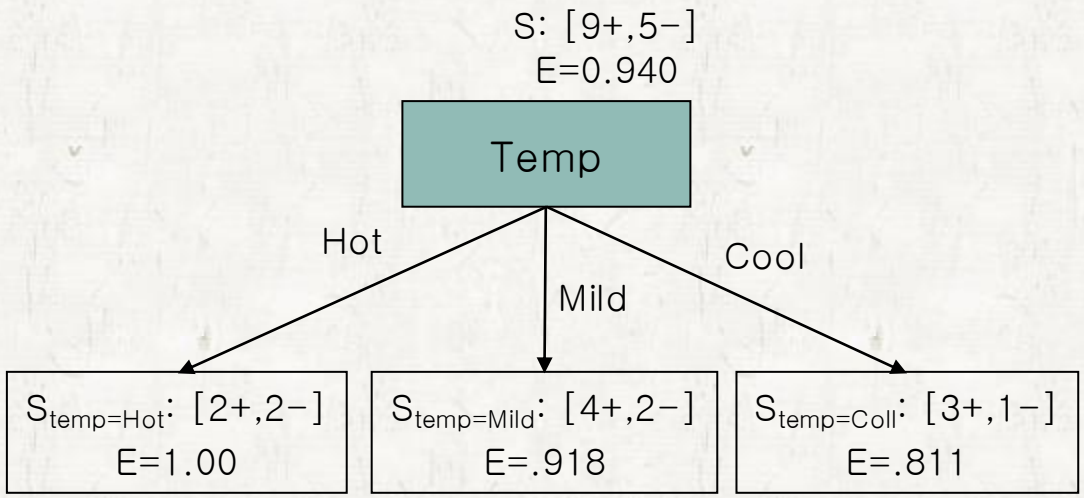
  - What is the gain after dividing with Humidity

$$Gain(S, Humidity) = 0.940 - 0.789 = 0.151$$

$$Gain(S,a) = Entropy(S) - \sum_{v \text{ that } a \text{ can take}} \frac{|S_{a=v}|}{|S|} Entropy(S_{a=v})$$

Selecting Attribute

- Which attribute is the best classifier?

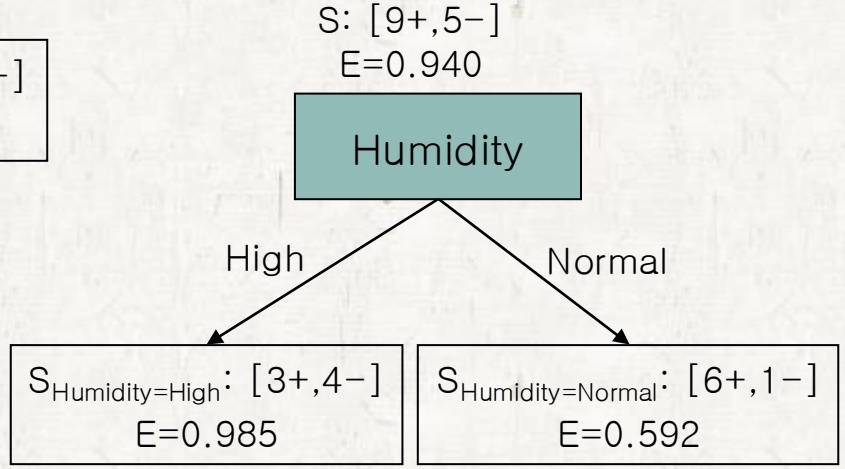


Entropy(S, Temp)
= (4/14)*1.0−(6/14)*.918−(4/14)*.811
=.911

Gain(S, Temp)
= .940−.911
=.029

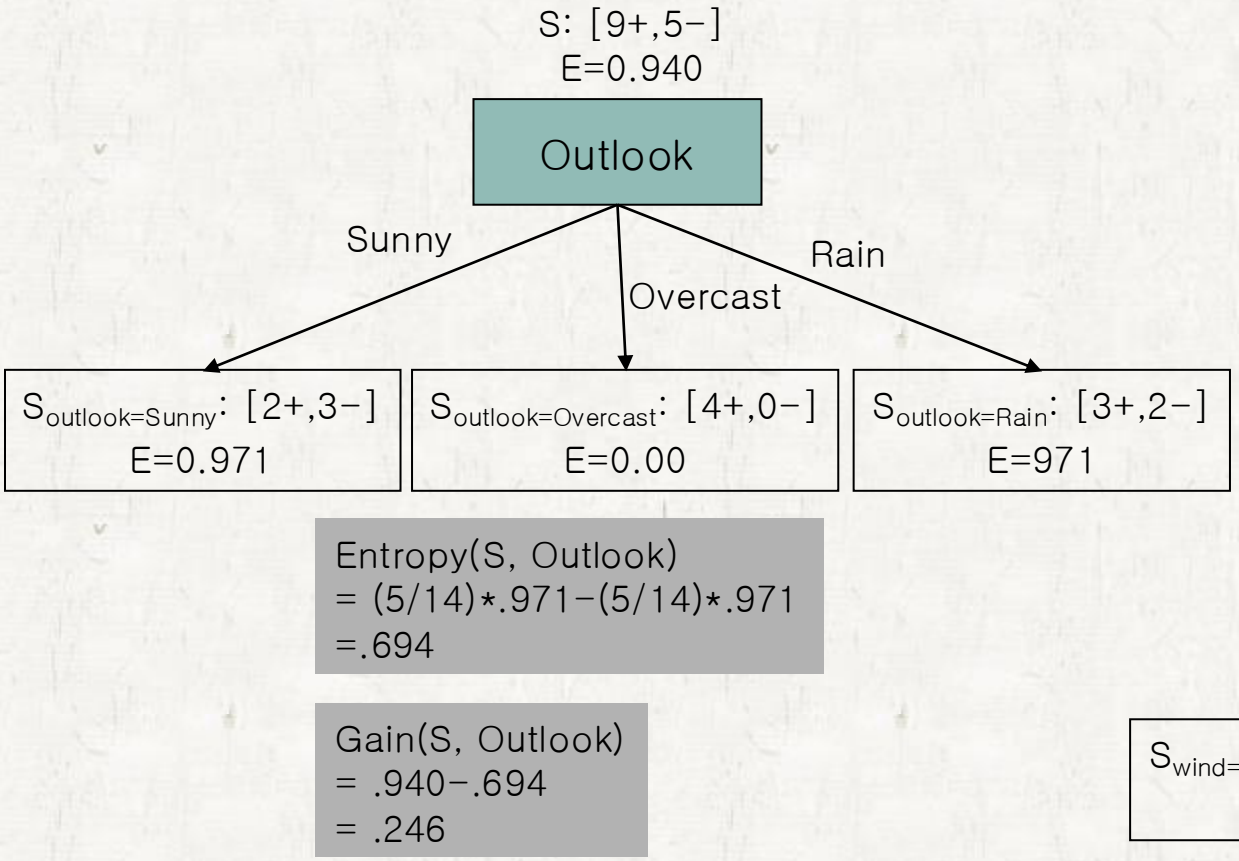
Entropy(S, Humidity)
= .(7/14)*.985−(7/14)*.592
= .798

Gain(S, Humidity)
= .940−.798
=.151



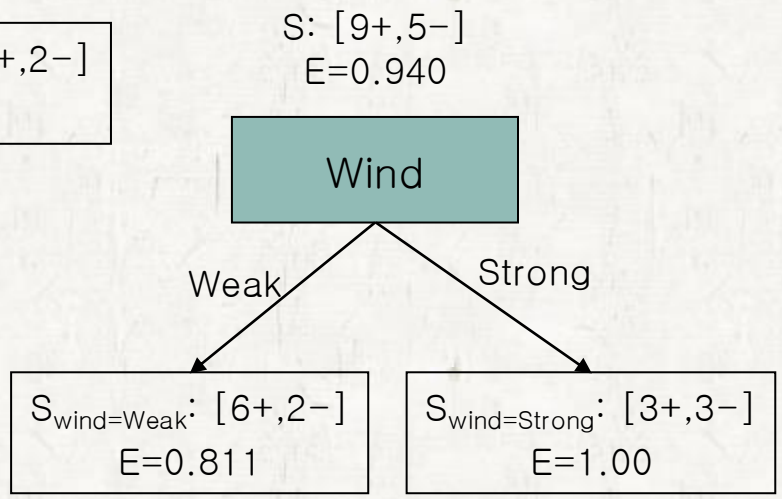
Selecting Attribute

- Which attribute is the best classifier?



Entropy(S, Wind)
= (8/14)*.811−(6/14)*1.00
=.892

Gain(S, Wind)
= .940−.892
=.048



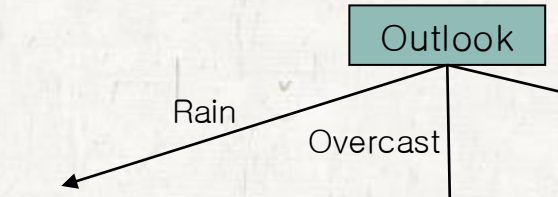
Selecting Attribute

- Which attribute is the best classifier?
  - At root node, we will classify the data with “Outlook”

Gain(S, Outlook) = 0.246
Gain(S, Humidity) = 0.151
Gain(S, Wind) = 0.048
Gain(S, Temp) = 0.029

Selecting Attribute

- Which attribute is the best classifier?
  - At root node, we will classify the data with “Outlook”



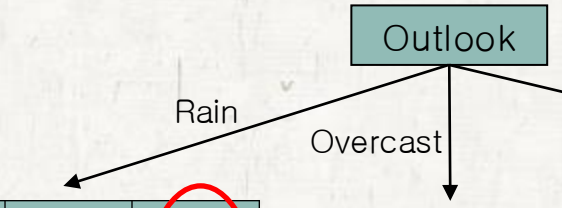
	Outlook	Temp	Humidity	Wind	Play
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
10	Rain	Mild	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes

	Outlook	Temp	Humidity	Wind	Play
3	Overcast	Hot	High	Weak	Yes
7	Overcast	Cool	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes

Selecting Attribute

- Which attribute is the best classifier?
  - At root node, we will classify the data with “Outlook”



	Outlook	Temp	Humidity	Wind	Play
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
10	Rain	Mild	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

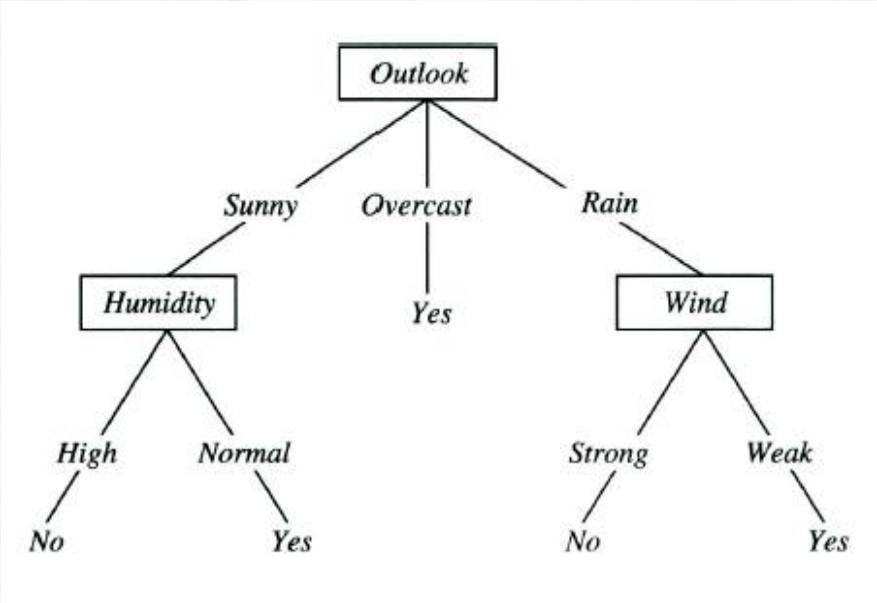
	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes

Yet, not completely separated. What will you do?

Recursively, apply the same procedure !!

Decision Tree Creation

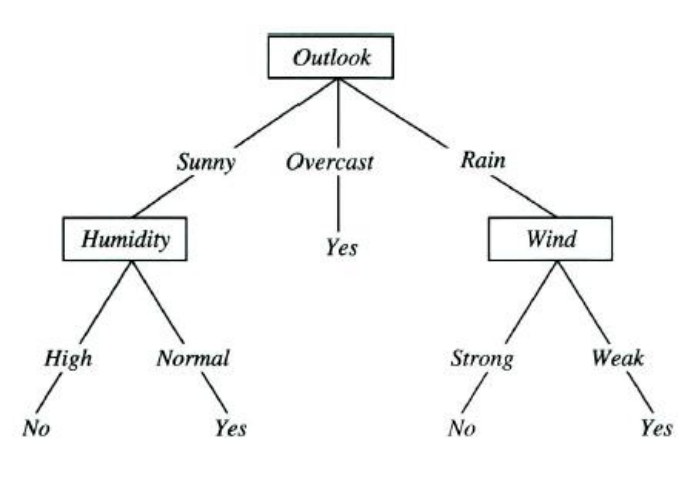
- Created Decision Tree



- Prediction
  - Sunny, Mild, High, Strong -> No

Decision Tree Creation

- Rule Generation from Decision Tree



If Outlook = Sunny & Humidity = High
then Tennis = No

If Outlook = Sunny & Humidity = Normal
then Tennis = Yes

If Outlook = Overcast
then Tennis = Yes

If Outlook = Rain & Wind = Strong
then Tennis = No

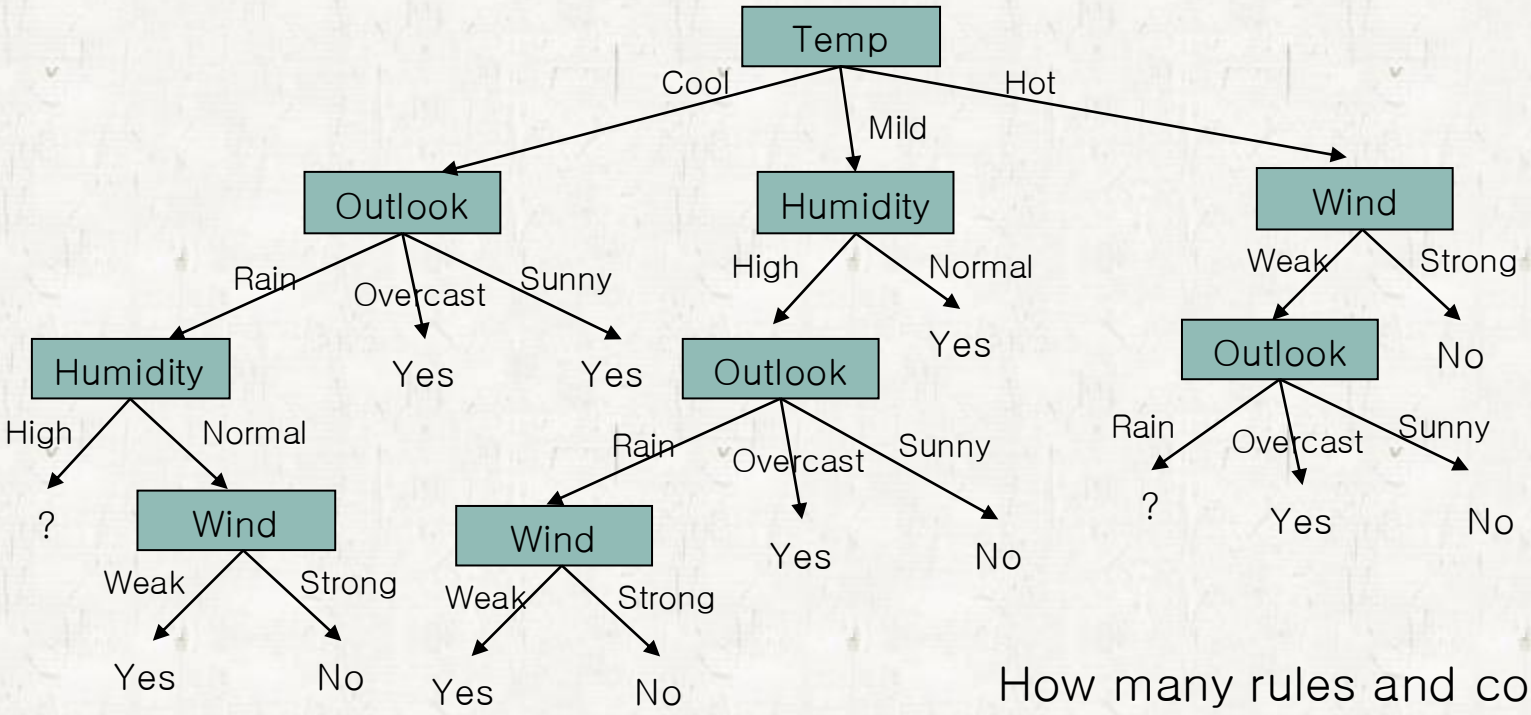
If Outlook = Sunny & Wind = Weak
then Tennis = Yes

Review of ID3

- Review of ID3
  - At each node, select the attribute which the best classifies the local training examples.
  - Trees that place high information gain attributes close to the root are preferred over those that do not
  - Shorter trees are likely to be preferred over longer trees
    => Generalization of the given data
    => Construct a decision tree from a root node by a greedy search process

Review of ID3

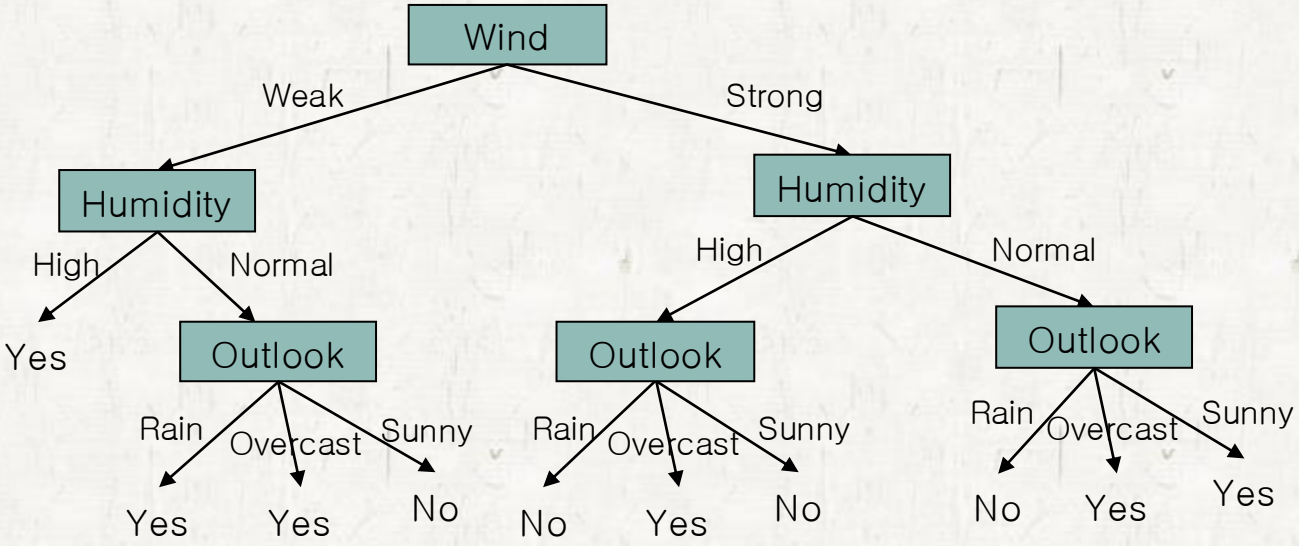
- Decision Tree by ID3 is not the unique decision tree for given data



How many rules and conditions?

Review of ID3

- Decision Tree by ID3 is not unique decision tree for given data



How many rules and conditions?

Review of ID3

- How to find the tree which has the minimum rules and conditions
  - At each node, choose the attribute which correctly split Yes/No as much as possible.

Greedy Algorithm!!